Module 5 Assignment: Variant Interpretation in CYP2C19

The ClinVar fetch returned 80 variants for CYP2C19, 38 labelled drug response, 33 VUS, 9 B/LB and no P/LP variants at all. Both anchor alleles arrived, confirming the fetch was complete, CYP2C19*2 (rs4244285, gnomAD AF 0.168) and CYP2C19*17 (rs12248560, AF 0.200).

1. **Thresholds and calibration**

   Frequency threshold MAX_AF = 0.25. A rare disease frequency cutoff was considered unsuitable because clinically relevant CYP2C19 alleles can be common. In this dataset, CYP2C19*2 had an allele frequency of 0.168 and *17 had a frequency of 0.200. The selected threshold retained both anchors and removed only 2 oof 80 variants, neither labelled P/LP nor drug response. Frequency was therefore used as a screening criterion rather than evidence that a variant lacks function.

   Effect of a rare disease cutoff, A threshold of 0.001 would have removed 19 variants, including 14 labelled drug response or P/LP. These included *2, *9, *3, and the regulatory variants rs12769205 and rs127777823. This shows how a strict rarity filter could exclude important pharmacogenetic evidence while retaining poorly characterized rare variants.

   Predictor selection, CADD was selected because it scored 68 of 80 variants, including noncoding and synonymous variants. REVEL and AlphaMissense each scored 54 variants, all missense. CADD therefore provided broader coverage of the variant types relevant to CYP2C19.

   Calibration limitation, after filtering, CYP2C19 contained 0 P/LP and 5 B/LB variants. Calibration produced one populated band with LR+ = 0.00 and four empty bands. This reflected the absence of positive examples, rather than strong benign evidence. Drug-response labels do not count as pathogenicity labels, leaving insufficient data for gene-specific calibration. CYP2C19 was therefore pooled with the other four pharmacogenes.

CADD band	n	n P/LP	n B/LB	LR+	ACMG evidence strength
0 – 10	72	1	71	0.03	strong, benign
10 – 20	33	0	33	0.00	strong, benign
20 – 25	17	8	9	2.17	supporting, pathogenic
25 – 30	6	4	2	4.89	moderate, pathogenic
30 – 50	34	34	0	inf	strong, pathogenic

Table 1. Pooled calibration of CADD across the five-gene pharmacogenes panel (616 variants, MAX_AF 0.25 applied throughout). Labelled variants contributed: DPYD 88 B/LB + 39 P/LP, VKORC1 11 + 7, CYP2C9 7 + 1, CYP2C19 5 + 0, TPMT 4 + 0.

The pooled calibration included 162 labelled variants, 47 P/LP and 115 B/LB. LR+ increased with CADD score, suggesting useful discrimination, but several limitations reduced confidence. The 25-30 band contained only six variants (4 P/LP and 2 B/LB), so its LR+ of 4.89 was considered unstable and the band was excluded from classification. The 30-50 band had an infinite LR+ because no benign variants were present, rather than because the evidence was strong.

Importantly, DPYD contributed 127 variants, while CYP2C19 contributed only five benign variants. The pooled estimates were therefore dominated by DPYD. Applying them to CYP2C19 assumes that CADD performs similarly across these different enzymes, which could not be tested with the available data. This limits confidence in the CYP2C19 classifications reported in Q5.

2. **Coverage gaps and the regulatory genome**

REVEL and AlphaMissense provided no scores for 26 of 80 CYP2C19 variants, including synonymous, upstream, structural, splice-donor, intronic, intergenic, stop-gained, and stop-loss variants. These fall outside the tools' missense scope. CADD covered more variant types, scoring six of seven synonymous and four of five upstream variants, but missed all eight structural deletions and the splice-donor variant.

An important unscored allele, rs72558186 (CYP2C19*7; c.819+2T>A) affects a canonical splice-donor site and is known non-functional allele associated with reduced drug metabolism. It received no CADD, REVEL, or AlphaMissense score. This demonstrates why missing scores must not be treated as benign evidence.

SpliceAI or Pangolin could provide additional predictions of donor disruption, although their scores would need to be calculated. Experimental confirmation could be obtained through a minigene assay or RT-PCR/RNA-seq in a suitable model. These experiments could identify abnormal transcripts and determine whether transcript abundance is reduced by nonsense-mediated decay.

The AlphaGenome analysis was skipped, leaving the regulatory assessment incomplete. This was particularly relevant to *7 and rs4244285 (CYP2C19*2). Although *2 was annotated as synonymous and had a CADD score of only 5.686, its functional effect involves abnormal splicing. Splicing attributes could have helped connect variants to their mechanisms, while regulatory attributes could have supported investigation of noncoding variants. However, no Atlas results were generated, so these mechanisms cannot be claimed as model findings. Conversation features alone would also leave the specific regulatory mechanism unresolved.

3. A discordant variant, read mechanistically

Variant rs41291556—CYP2C19*8, p.Trp120Arg (W120R), triggered `flag_revel_vs_am` because REVEL and AlphaMissense differed by 0.50, exceeding the 0.4 threshold.

Tool / field	Value	Reads as
CADD phred	25.70	top ~0.3% of genome-wide deleteriousness
REVEL	0.440	below the usual 0.5 pathogenicity threshold: tolerated
AlphaMissense	0.9402	likely pathogenic
gnomAD AF	0.001571	rare
ClinVar	drug response	functionally characterized, not a P/LP assertion

Table 2. Tool outputs for rs41291556

A possible explanation is that the tools capture different aspects of variant effects, REVEL was developed to distinguish disease-associated missense variants from neutral variants, which may not fully represent changes in drug metabolism. The replacement of tryptophan with a charged arginine could affect protein function, consistent with the higher AlphaMissense and CADD scores. However, the scores alone cannot establish the structural mechanism or whether enzyme activity is partially or completely lost.

Similar disagreements were observed for *R186P (22) and **F448L (*25)**, with REVEL scores of 0.209 and 0.254 versus AlphaMissense scores 0.800 and 0.798. This repeated pattern suggests a possible calibration difference, but does not prove that training labels caused the disagreement or exclude other explanations. Functional assays and curated allele-function evidence are therefore needed. These findings support avoiding reliance on REVEL alone, CADD was retained primarily because of its broader variant coverage.

4. What this analysis cannot establish

The pipeline contains population allele frequencies rather than invidual genotypes, so CYP2C19 diplotypes and metabolizer phenotypes could not be determined. It also could not establish whether W120R and other variants, such as *17, occur on the same chromosome. These limitations prevent patient-specific interpretation.

Co-medications, liver function, enzyme expression, and inflammation can modify CYP2C19 activity and cause phenoconversion. The provisional findings therefore describe possible variant effects rather than an individual's drug response. ClinVar drug-response labels also do not establish how often carriers experience a clinical outcome.

A quantitative enzyme assay comparing W120R with wild-type CYP2C19, using relevant substrates and LC-MS measurements, would help establish the direction and magnitude of its effect. Deep mutational scanning could extend testing to additional coding variants, but would not resolve phase, regulatory variants, or patient-level modifiers. Near-normal activity would weaken the loss-of-function interpretation, while substantially reduced activity would support it. However, assay results alone would not prove why the predictors disagreed or automatically determine clinical recommendations.

Eight VUS shifted toward pathogenic and 16 toward benign under the provisional classification rules. The pathogenic shifts relied on the CADD 20-25 band, using pooled calibration dominated by DPYD. These shifts represent limited computational evidence and should not be presented as clinical reclassifications. A further six VUS, all large structural deletions, remained unscored and unresolved.

AI Use Disclosure

Claude (Anthropic, 2026) was used to structure the report and to mitigate errors. All content and analysis are my own. Anthropic. (2026). Claude (Opus 4.5) [Large language model]. https://claude.ai